

*MA.8.AR.2.2***Benchmark**

MA.8.AR.2.2 Solve two-step linear inequalities in one variable and represent solutions algebraically and graphically.

Benchmark Clarifications:

Clarification 1: Instruction includes inequalities in the forms $px \pm q > r$ and $p(x \pm q) > r$, where p , q and r are specific rational numbers and where any inequality symbol can be represented.

Clarification 2: Problems include inequalities where the variable may be on either side of the inequality.

Connecting Benchmarks/Horizontal Alignment

- MA.8.AR.1.3
- MA.8.GR.1.3

Terms from the K-12 Glossary

- Coefficient
- Linear expression
- Rational Number

Vertical Alignment**Previous Benchmarks**

- MA.7.AR.2.1

Next Benchmarks

MA.912.AR.2.6

Purpose and Instructional Strategies

In grade 7, students wrote and solved one-step inequalities in one variable within a mathematical context and represented solutions algebraically or graphically. In grade 8, students solve two-step linear inequalities in one variable and represent solutions algebraically and graphically. In Algebra 1, students will extend this learning to write and solve one-variable linear inequalities, including compound inequalities representing solutions algebraically or graphically. Additionally, students will write and solve two-variable linear inequalities to represent relationships between quantities from a graph or a written description within a mathematical or real-world context.

- Instruction emphasizes the properties of inequality with connections to the properties of equality (*MTR.5.1*).
- Instruction includes showing why the inequality symbol reverses when multiplying or dividing both sides of an inequality by a negative number.
 - For example, if the inequality $6 > -7$ is multiplied by -3 , it results in $-18 > 21$ which is a false statement; therefore, the inequality symbol must be reversed to keep a true statement. Since 6 is to the right of -7 on the number line and multiplying by a negative number reverses directions, $6(-3)$ will be to the left of $-7(-3)$ on the number line.
- Instruction includes cases where the variable is on the right side of the inequality.
- Variables are not limited to x ; instruction includes using a variety of lowercase letters for their variables; however, the letters o , i , and l should be avoided as they too closely resemble zero and one.
- Instruction emphasizes the understanding of defining an algebraic inequality. Students

should have practice with inequalities in the form of $px \pm q > r$, $px \pm q < r$, $px \pm q \geq r$ and $px \pm q \leq r$. Students should explore how "is greater than or equal to" and "is strictly greater than" are similar and different as well as "is less than or equal to" and "is strictly less than." Students should use academic language when describing an algebraic inequality.

Common Misconceptions or Errors

- Students may confuse when to use an open versus closed circle when graphing an inequality. Emphasize the inclusion ($\leq$ and $\geq$) versus non-inclusion ($<$ and $>$) of that value as a viable solution and provide problems that motivate reasoning with different ranges of possible values for the variable.
- Some students are unable to see the difference between the multiplication or division property of equality and the multiplication or division property of inequality.
- Students may misunderstand the direction the inequality symbol is pointing is always the direction they shade on the number line. To address this misconception, emphasize reading the inequality sentence aloud and use numerical examples to test for viable solutions (*MTR.6.1*).

Strategies to Support Tiered Instruction

- Instruction includes the use of real-world inequality problems to help students determine when to use an open versus closed circle when graphing an inequality. Teacher facilitates discussion around whether various solutions make sense by having students graph all possible solutions on a number line and then deciding if the solutions make sense in the context of the problem.
 - For example, Henry has up to \$20 to spend at the football game and the dance after the game. He must buy a dance ticket for \$13 and can spend the rest on hot dog and drink combinations at the football game for \$2 per combo. After Henry buys his dance ticket, how many hot dog and drink combinations could Henry purchase? (The c represents the hotdog and drink combo.)
Students can write the inequality $2c + 13 \leq 20$ to represent the situation.
Students should get the algebraic solution as $c \leq 3.5$, however, within the context of the problem the possible solutions are 0, 1, 2 or 3.
- Teacher models solving an equation and its corresponding inequality. Teacher facilitates discussion about their similarities and differences, paying close attention to cases when multiplying or dividing with negative values and using substitution to verify the solutions.
- Instruction includes using substitution to test possible solutions to determine the correct direction to shade on the number line.
- Teacher provides opportunities for students to use manipulatives for solving inequalities. When using manipulatives, ensure students use the appropriate inequality symbol, rather than an equal sign.
- Instruction includes emphasizing reading the inequality sentence aloud and use numerical examples to test for viable solutions.

Instructional Tasks

Instructional Task 1 (MTR.7.1)

As a social media employee, Rick is paid \$100 a week plus \$5 for every person that he adds to the website. This week, Rick wants his pay to be at least \$200.

Part A. Write and solve an inequality for the number of sales Rick needs to make.

Part B. Graph the solution on a number line.

Part C. Describe what the solutions mean within the context of the problem.

Instructional Task 2 (MTR.4.1, MTR.7.1)

At his job, Jake earns \$7.50 per hour. He also earns a \$55 bonus every month. Jake needs to earn at least \$235 every month.

Part A. Jake determines that the inequality $7.50h + 55 \geq 235$ can be used to calculate the number of hours he needs to work each month. Explain why the symbol $\geq$ is used within the inequality.

Part B. Solve and graph the solution set for the number of hours Jake needs to work each month.

Instructional Items

Instructional Item 1

Represent the solutions to the inequality $550 + 8b > 925$ graphically.

Instructional Item 2

Solve and graph the inequality $-225 \leq 320 - 0.5x$.

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*